

Skills Summary

Solving Exponential and Logarithmic Equations

Use this reference while completing your worksheet or when studying.

Skill 1 — Match the bases

Rule: if $b^M = b^N$ with $b > 0$, $b \neq 1$, then $M = N$. Rewrite both sides as powers of one base, then set the exponents equal. This is the exact method — use it whenever the numbers are powers of a common base.

Example: Solve $2^{3x} = 32$.

Step 1. The right side is a power of the same base as the left, so no logarithm is needed — matching the bases gives an exact answer.

Step 2. $32 = 2^5$, so $2^{3x} = 2^5$ and therefore $3x = 5$.
 $\Rightarrow x = \frac{5}{3}$

Try it: Solve $4^{x-1} = 64$.

check: $x = \frac{5}{3}$

Skill 2 — Take a logarithm of both sides

Method: when the two sides are not powers of a common base, apply a logarithm to both sides and use the power rule to bring the unknown down:

$$b^x = c \implies x \ln b = \ln c \implies x = \frac{\ln c}{\ln b}.$$

Example: Solve $5^x = 40$, to two decimal places.

Step 1. 40 is not a power of 5, so matching bases is impossible and a logarithm is the tool that frees an exponent.

Step 2. $x \ln 5 = \ln 40$, so $x = \frac{\ln 40}{\ln 5} = \frac{3.68887\dots}{1.60943\dots} = 2.29202\dots$
 $\Rightarrow 2.29$

Try it: Solve $3^x = 50$, to two decimal places.

check: $x = 2.29$

Skill 3 — Logarithmic equations

Method: condense to a single logarithm if there is more than one, then rewrite in exponential form and solve. **Always finish with a domain check:** every logarithm's input must be positive at your answer, or the candidate is discarded.

Example: Solve $\log_3(x - 4) = 3$.

Step 1. The unknown is locked inside the logarithm, so rewriting the statement as a power is what releases it.

Step 2. $3^3 = x - 4$, so $27 = x - 4$. The input at $x = 31$ is $27 > 0$, so the answer survives the domain check.

$$\Rightarrow x = 31$$

Try it: Solve $\log_2(x + 5) = 4$.

check: $11 = x$

Skill 4 — Growth and decay models

Shape: $A = A_0 \cdot r^t$, where A_0 is the starting amount and r is the factor per period ($r > 1$ grows, $0 < r < 1$ decays). To find an *amount*, substitute t . To find a *time*, isolate the power and take logarithms.

Example: A town of 8000 people grows by 3% a year, so $P = 8000(1.03)^t$. Find the population after 12 years.

Step 1. The unknown here is the amount, not the time, so the model only has to be evaluated — no logarithm is involved.

Step 2. $1.03^{12} = 1.425760\dots$, and $8000 \times 1.425760\dots = 11406.087\dots$

$$\Rightarrow 11406.09$$

Try it: An account of 5000 dollars grows by 4% a year: $A = 5000(1.04)^t$. Find its value after 10 years, to the nearest cent.

check: 7401.22